

TANNAS, Sweden

WMO: 023080

Lat: 62.449N

Lon: 12.668E

Elev: 724

StdP: 92.92

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -21.3	(c) -18.5	(d) -24.0	(e) 0.5	(f) -20.9	(g) -21.1	(h) 0.6	(i) -18.0	(j) 6.0	(k) -0.6	(l) 5.3	(m) -1.3	(n) 0.3	(o) 120	(p) 0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	22.9	14.8	20.9	13.7	18.9	12.8	16.0	20.7	14.8	18.8	13.8	17.5	1.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.2	11.0	17.2	13.1	10.3	16.2	12.1	9.6	15.3	47.1	20.7	43.9	18.9	40.9	17.6	19.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 4.8	(b) 4.1	(c) 3.4	(e) -23.7	(f) 25.5	(g) 2.8	(h) 1.9	(i) -25.6	(j) 26.9	(k) -27.3	(l) 28.0	(m) -28.8	(n) 29.1	(o) -30.8	(p) 30.5		
(a) 4.8	(b) 4.1	(c) 3.4	(e) -24.0	(f) 17.6	(g) 2.7	(h) 1.2	(i) -26.0	(j) 18.5	(k) -27.5	(l) 19.2	(m) -29.0	(n) 19.9	(o) -31.0	(p) 20.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.8	-7.3	-6.6	-4.2	0.6	5.6	9.6	12.7	11.4	7.4	1.7	-3.4	-6.3
	DBStd	8.23	5.13	5.31	4.54	3.60	4.20	3.69	3.31	3.16	3.22	3.60	4.47	5.38
	HDD10.0	3212	536	465	440	282	149	52	11	22	90	259	402	504
	HDD18.3	6033	794	698	698	532	394	263	177	216	328	517	652	763
	CDD10.0	224	0	0	0	0	13	39	95	64	12	1	0	0
	CDD18.3	3	0	0	0	0	0	0	2	1	0	0	0	0
	CDH23.3	43	0	0	0	0	3	7	29	4	0	0	0	0
	CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0
Wind	WSAvg	1.4	1.4	1.5	1.7	1.5	1.6	1.4	1.2	1.1	1.3	1.4	1.3	1.4
Precipitation	PrecAvg	628	42	31	30	32	54	73	90	89	52	52	47	36
	PrecMax	769	90	53	49	84	107	134	168	138	89	128	115	56
	PrecMin	489	9	5	9	6	14	17	29	40	16	14	21	11
	PrecStd	74	20	11	10	18	25	30	33	30	23	30	20	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.3	4.2	7.0	12.9	23.1	24.4	26.5	23.7	19.3	12.7	7.6	4.3
		MCWB	1.4	1.7	2.9	6.7	13.0	13.8	17.1	15.4	12.4	8.9	6.3	3.1
	2%	DB	2.1	2.7	5.0	10.2	19.2	21.4	23.5	21.3	16.4	9.9	5.3	3.0
		MCWB	0.8	0.8	2.2	5.0	11.0	12.7	15.7	14.3	11.7	7.5	4.2	1.9
	5%	DB	0.8	1.5	3.2	8.0	16.0	18.8	21.3	19.1	14.1	8.3	3.4	1.8
		MCWB	-0.2	0.0	0.9	3.8	9.5	11.5	14.7	13.7	10.6	6.6	2.5	0.8
	10%	DB	-0.5	0.2	1.9	6.0	12.8	16.4	19.1	16.9	12.3	6.8	2.0	0.6
		MCWB	-1.3	-1.0	0.1	2.7	7.6	10.8	13.7	12.9	9.8	5.4	1.2	-0.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.8	2.4	3.7	7.8	13.6	15.4	18.0	16.7	13.7	10.0	6.3	3.5
		MCDB	2.8	3.8	6.1	11.7	20.5	20.3	23.9	21.4	17.1	11.4	7.4	4.1
	2%	WB	0.9	1.2	2.3	5.5	11.6	13.6	16.6	15.4	12.3	8.1	4.3	2.0
		MCDB	1.8	2.3	4.5	9.2	17.8	19.3	21.9	19.0	15.2	9.4	5.2	2.8
	5%	WB	-0.1	0.1	1.2	4.2	9.9	12.5	15.4	14.4	11.1	6.7	2.6	1.0
		MCDB	0.7	1.4	3.0	7.2	14.8	17.6	19.9	17.8	13.5	8.0	3.2	1.7
	10%	WB	-1.3	-1.1	0.1	3.1	8.2	11.3	14.4	13.3	10.0	5.6	1.3	-0.3
		MCDB	-0.5	0.2	1.7	5.6	12.4	15.6	18.2	16.4	11.9	6.7	1.8	0.5
Mean Daily Temperature Range	5% DB	MDBR	5.1	5.8	7.2	7.3	9.2	9.4	9.5	9.0	7.6	5.5	4.4	4.9
		MCDBR	5.5	5.7	7.5	9.7	14.3	13.5	13.6	12.5	10.7	8.1	5.3	5.0
		MCWBR	4.9	4.7	5.3	5.7	7.3	6.7	6.8	6.5	6.3	5.6	4.7	4.7
	5% WB	MCDBR	5.2	5.3	6.9	8.8	13.0	11.5	11.8	10.5	9.5	7.0	5.1	5.0
		MCWBR	4.8	4.6	5.1	5.4	7.1	6.2	6.3	6.1	6.0	5.2	4.7	4.9
Clear-Sky Solar Irradiance	taub	0.194	0.250	0.279	0.311	0.323	0.319	0.332	0.322	0.294	0.277	0.207	0.170	
	taud	2.065	2.249	2.260	2.314	2.466	2.516	2.487	2.504	2.567	2.479	2.134	1.971	
	Ebn at Noon	510	728	835	878	898	907	883	864	831	702	481	326	
	Edh at Noon	32	59	86	101	95	93	93	84	65	48	29	18	
All-Sky Solar Radiation	RadAvg	0.23	0.84	2.33	4.17	4.76	5.08	4.65	3.59	2.29	1.00	0.31	0.12	
	RadStd	0.02	0.13	0.20	0.35	0.35	0.44	0.56	0.33	0.26	0.15	0.06	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page